

Zhiyuan Peng

RESEARCH SCIENTIST · APPLIED SCIENTIST · DATA SCIENTIST · MACHINE LEARNING ENGINEERING

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Work Experience

Walmart Global Tech

Sunnyvale, California

DATA SCIENTIST INTERN

Jun. 2022 - Sep. 2022

- Developed an entity-aware multi-task learning model to improve query understanding, **reducing GPU resource usage by 75% and enhancing development and maintenance efficiency.**
- Conducted online A/B testing that achieved statistically significant improvements over existing production metrics: **Gross Merchandise Volume by 0.51%, Order Count by 0.65%, Units by 1.08%, Add-to-Cart by 0.65%, and P99 Latency by 18.6%.**
- Published a paper at KDD'23 titled **Entity-aware Multi-task Learning for Query Understanding at Walmart.**

Aktana. Co., Ltd.

San Francisco, California

MACHINE LEARNING ENGINEERING INTERN

Jun. 2020 - Dec. 2020

- Addressed the lack of evaluation data by manually tagging approximately 300 call notes with NER labels using the Brat rapid annotation tool.
- Developed an intent classification model to predict doctors' intents and a named entity recognition model for extracting valuable information from call notes taken by pharmaceutical sales representatives.
- Created a demonstration app for recommendation suggestions for pharmaceutical sales representatives, implemented using Flask, and deployed on AWS EC2.

Research Projects

ELOQ: Resources for Enhancing LLM Detection of Out-of-Scope Questions

Santa Clara, California

SANTA CLARA UNIVERSITY

Aug. 2024 - Present

- Designed, implemented, and evaluated a novel method to efficiently generate borderline out-of-scope questions for a set of provided documents.
- Demonstrated usefulness of our synthetic data by training detectors for out-of-scope questions.
- Empirically evaluate several LLMs as RAG agents for accuracy in detecting confusion and generating appropriate responses.
- Paper **ELOQ: Resources for Enhancing LLM Detection of Out-of-Scope Questions** was accepted by SIGIR 2025.

Soft Prompt Tuning for Augmenting Dense Retrieval with Large Language Models

Santa Clara, California

SANTA CLARA UNIVERSITY

May. 2023 - July. 2023

- Proposed the first application of soft prompt tuning in conjunction with large language models to enhance dense retrieval (DR) tasks.
- Developed a soft prompt filter to refine document-query pair selection within prompts, significantly improving the quality of weak data used for training DR models.
- Implemented a BM25-based post-filtering technique to minimize noise in the training data, substantially boosting retrieval performance.
- Published a paper in Knowledge-Based Systems (Impact Factor 8.8 in 2023), available at **Soft Prompt Tuning for Augmenting Dense Retrieval with Large Language Models.**

Education

Santa Clara University

California, U.S.A

DOCTORATE IN COMPUTER SCIENCE AND ENGINEERING

Jan. 2021 - Jan/Mar. 2026 (Expected)

- Advised by prof. Yi Fang to do research in enhancing search systems with large language models
- NAACL 2025 Student Volunteer Award**
- KDD 2023 Student Travel Award**

Santa Clara University

California, U.S.A

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Fall. 2018 - Dec. 2020

- Thesis **"Classification and Named Entity Recognition of Short Text"** advised by prof. Yi Fang

Beijing Institute of Technology

Beijing, China

M.S. IN INFORMATION AND COMMUNICATION ENGINEERING

Sep. 2014 - Mar. 2018

- Academic Scholarship winner, Beijing Institute of Technology, 2015

- Excellent Research Prize (3/589), School of Electrical and Information Engineering, Beijing Jiaotong University, 2014

Publications

SELECTED ARTICLES

ELOQ: Resources for Enhancing LLM Detection of Out-of-Scope Questions

Zhiyuan Peng, Jinming Nian, Alexandre Evfimievski, Yi Fang

The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2025

Soft prompt tuning for augmenting dense retrieval with large language models

Zhiyuan Peng, Xuyang Wu, Qifan Wang, Yi Fang

Knowledge-Based Systems 309 (2025) p. 112758. Elsevier, 2025

Entity-aware multi-task learning for query understanding at walmart

Zhiyuan Peng, Vachik Dave, Nicole McNabb, Rahul Sharnagat, Alessandro Magnani, Ciya Liao, Yi Fang, Sravanthi Rajanala

Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023

JOURNAL ARTICLES

Semi-supervised named entity recognition with data augmentation by structured consistency training

Zhiyuan Peng, Behnoush Abdollahi, Min Xie, Yi Fang

Expert Systems with Applications (2025) p. 127464. Pergamon, 2025

Multiple-access interference mitigation for acquisition of code-division multiple-access continuous-wave signals

Yuyao Shen, Yongqing Wang, Zhiyuan Peng, Siliang Wu

IEEE Communications Letters 21.1 (2016) pp. 192–195. IEEE, 2016

CONFERENCE PROCEEDINGS

Evaluating and enhancing large language models for novelty assessment in scholarly publications

Ethan Lin, Zhiyuan Peng, Yi Fang

AI & Scientific Discovery Workshop NAACL 2025 workshop, 2025

Passage-specific Prompt Tuning for Passage Reranking in Question Answering with Large Language Models

Xuyang Wu, Zhiyuan Peng, Krishna Sravanthi Rajanala Sai, Hsin-Tai Wu, Yi Fang

The Second Workshop on Generative Information Retrieval, 2024

DIANES: a DEI audit toolkit for news sources

Xiaoxiao Shang, Zhiyuan Peng, Qiming Yuan, Sabiq Khan, Lauren Xie, Yi Fang, Subramaniam Vincent

Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022

Multi-label classification of short texts with label correlated recurrent neural networks

Zhiyuan Peng, Behnoush Abdollahi, Min Xie, Yi Fang

Proceedings of the 2021 ACM SIGIR international conference on theory of information retrieval, 2021

High-precision TT&C signal simulation technology based on lagrange interpolation in high dynamic environment

Shao-Zhong Lu, Sheng Li, Zhi-Yuan Peng, Zhen-Gang Tang

2016 13th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), 2016

PREPRINTS

W-RAG: Weakly Supervised Dense Retrieval in RAG for Open-domain Question Answering

Jinming Nian, Zhiyuan Peng, Qifan Wang, Yi Fang

arXiv preprint arXiv:2408.08444 (2024). 2024

Q-peft: Query-dependent parameter efficient fine-tuning for text reranking with large language models

Zhiyuan Peng, Xuyang Wu, Qifan Wang, Sravanthi Rajanala, Yi Fang

arXiv preprint arXiv:2404.04522 (2024). 2024

Program Committees & Reviewers

Conferences ICLR, SIGIR, WWW, ACL, EMNLP, NAACL, COLING, KDD

Journals Information Sciences, Data Science and Management